STi7109 SoC + Windows CE Enables Connected High-Def Devices

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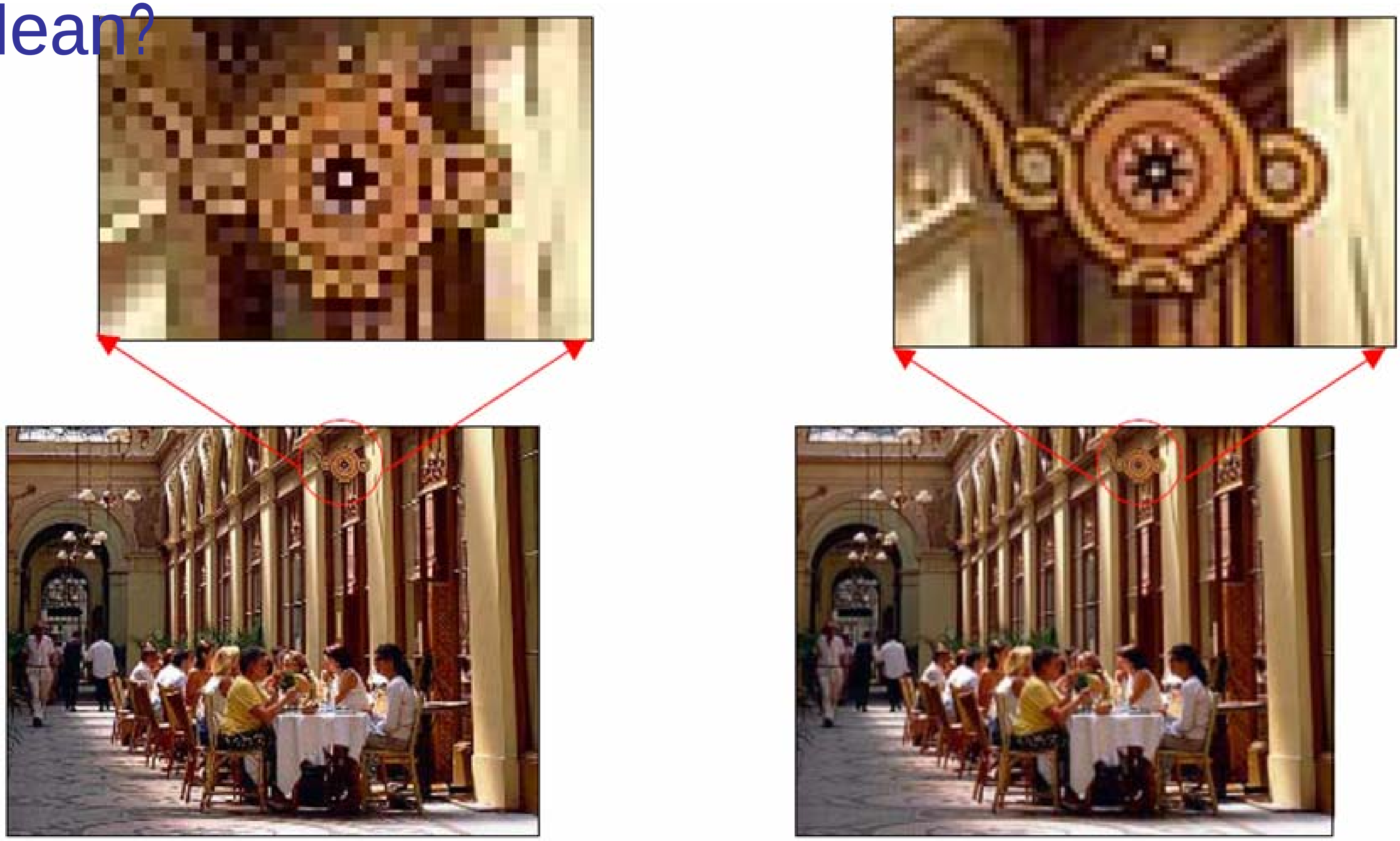
The World We Live In

Our High Definition World

- **For 2006, iSupply forecasts:**
  - 20 Mpcs HD STB
  - ~200 Mpcs HDTV
- **By 2010, 70% of all US TV sales will be HD**

What Does HD, High Definition “really” Mean?



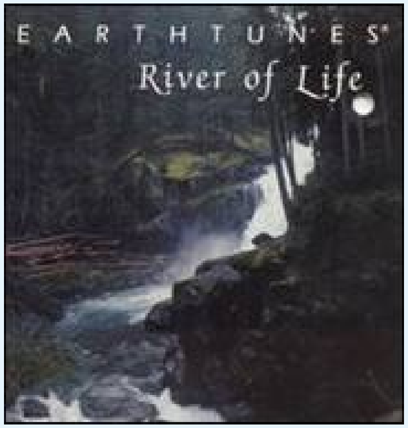
Standard Definition : 720x576
400K pixels

High Definition : 1920x1080
Two millions pixels

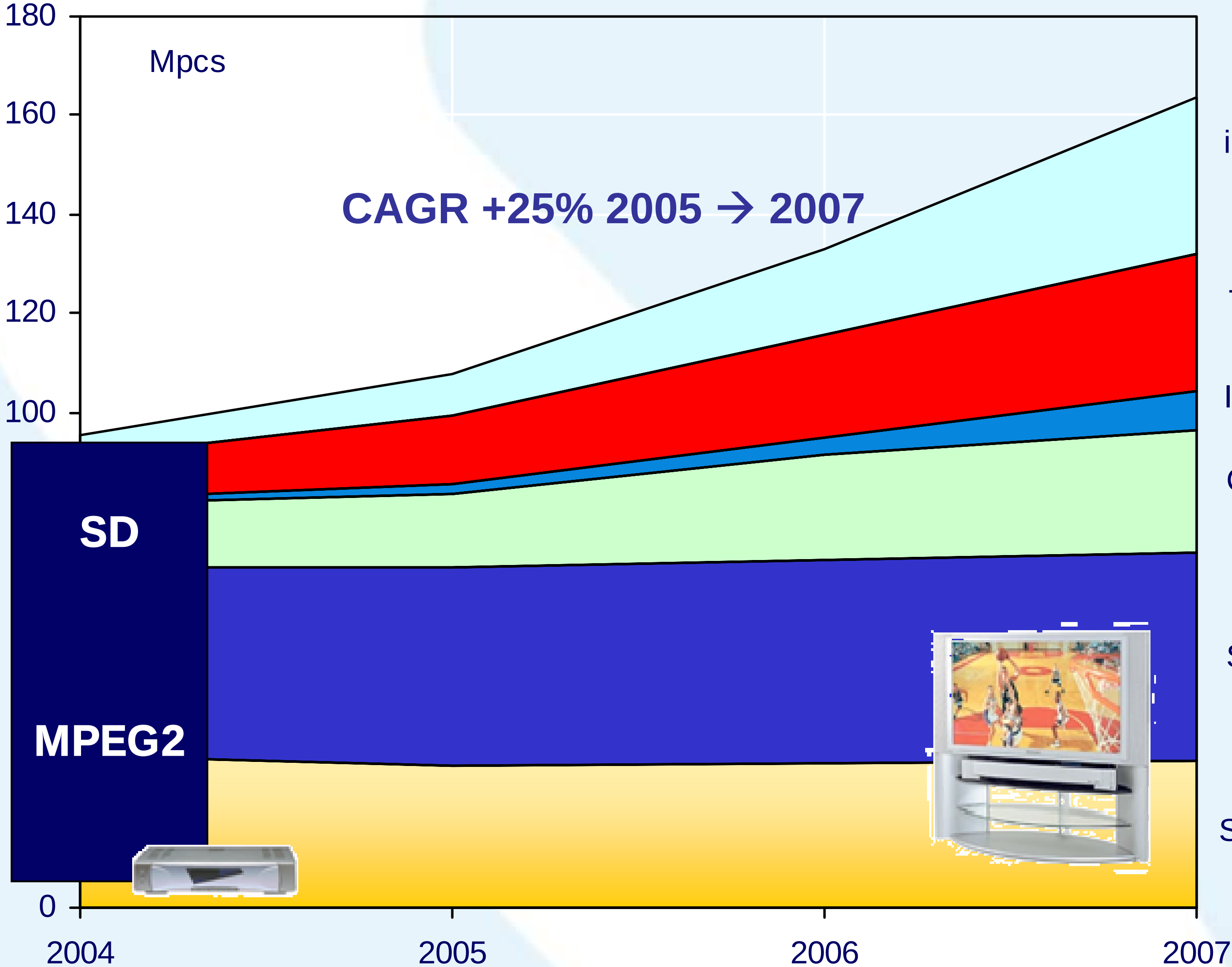
Our Digital Media World

- **20 Million homes in the US are labeled as a “digital home,” consisting of a broadband connection, a home network, and multiple networked devices**
- **Amount of digital content stored in that US digital home is increasing from 322 GB in 2005 to 1933 GB in 2010**

Source: The Diffusion Group, *Trends in Digital Home Storage*



Our Encoded World



HD

DVR

VC-1 / MPEG4

IP

Emerging Markets

* Source : inStat 2005, ST Sales & Market Research, Feb. 2006

Consequences of Advanced Codecs

Memory footprint

- MPEG2 requires 9 MB for HDTV (2 references + 1 frame decode)
- H264 requires more than 15 MB (3 MB for CPB, 12 MB for DPB)

=> balanced by SDRAM roadmap going to higher density for same cost

Memory bandwidth

- HDTV H264 decoder requires more than twice the bandwidth needed for MPEG2
- SDTV is even worse since more motion vector per macroblocks are allowed

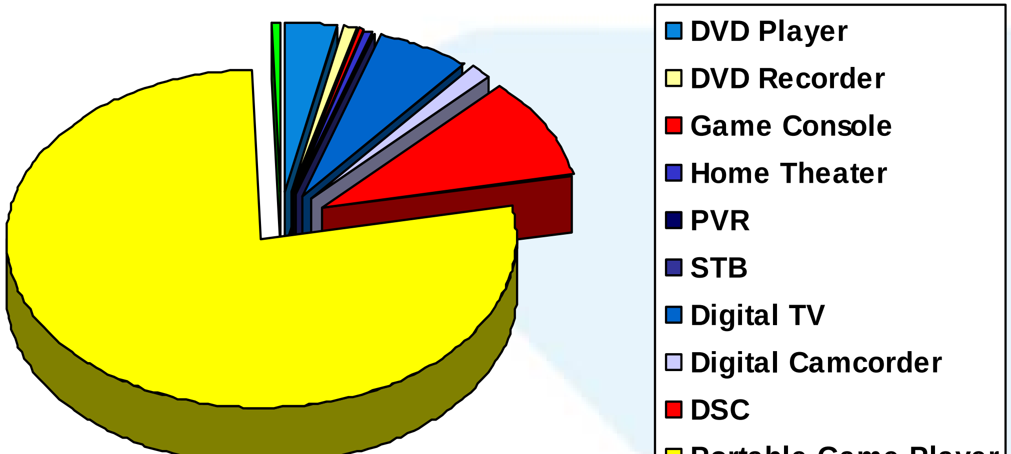
=> balanced by SDRAM roadmap going to higher frequencies

Processing power

- A software implementation of H264 decoder requires 6x more processing than MPEG2

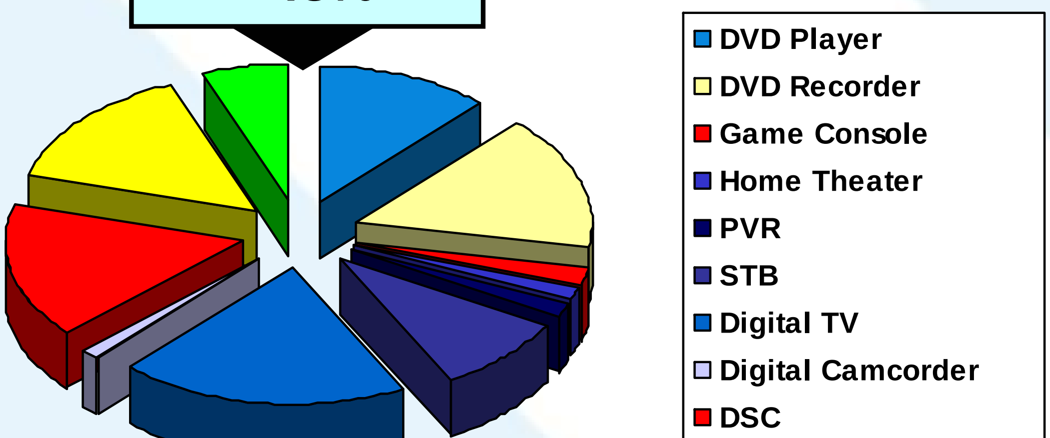
=> Simply not achievable in pure software at Consumer Electronics cost or power consumption

Our Connected World



Y2006: total 31Mpcs
7% of appliances

CAGR
+48%



Y2010: total 150Mpcs
24% of appliances

Connectivity in STB:

- ***USB and Ethernet*** *on all platforms*
- *Dish deploying* ***homeplug***
- *US cable :* ***MoCA / HPNA***
- *802.11g/a: disappointing,*
  =>now focusing on ***802.11n***

⇒***But none good enough…yet!***

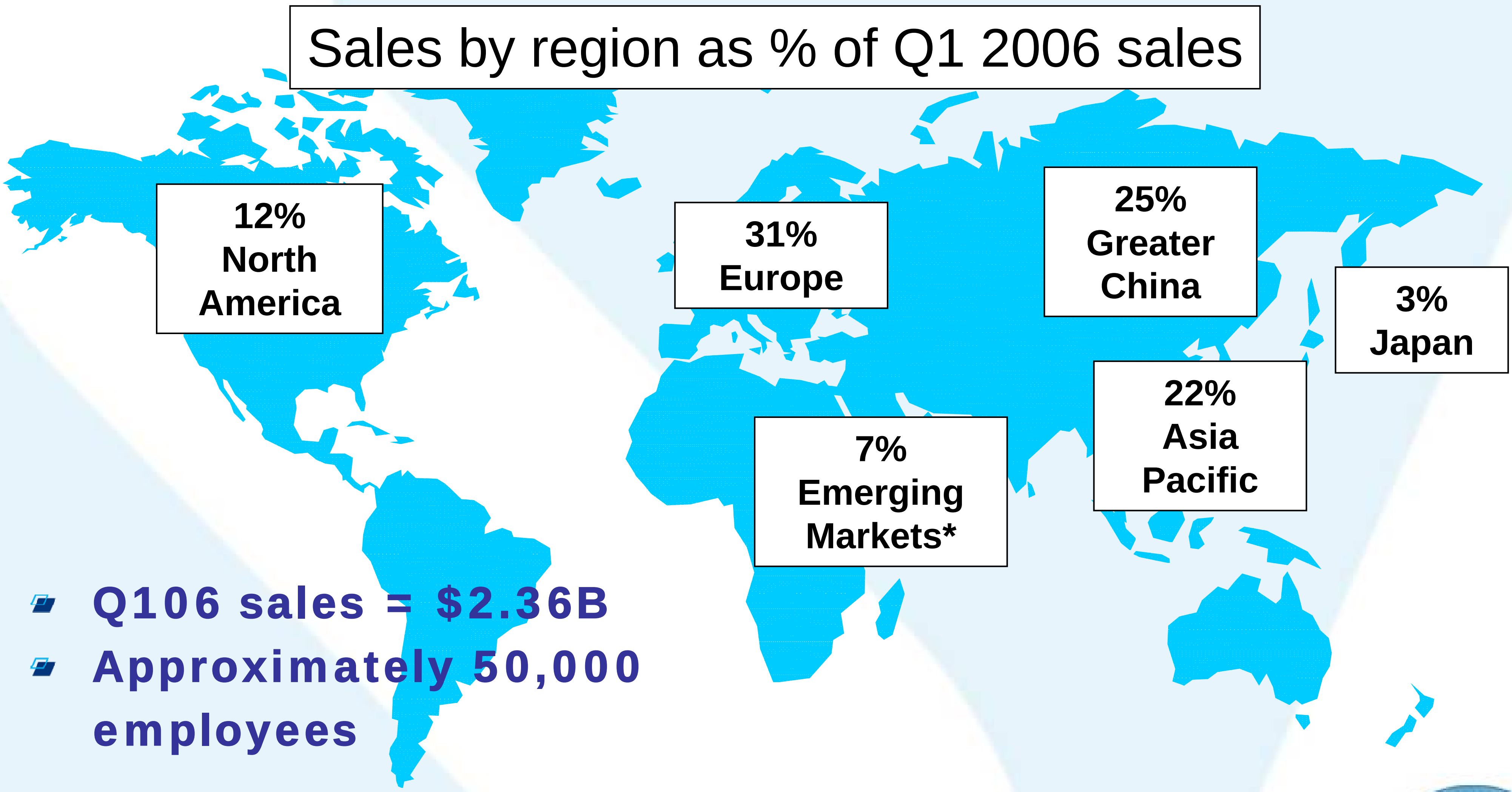
Source: ABI Research Q405

Imagine what you could do...

- **With a platform that has:**
  - HD video and graphics output to drive all of the HDTV's being bought
  - Advanced VC-1 and H.264 codec hardware exposed through a standard API
  - Connectivity hardware and protocols to get at all your digital media
- **What if you could quickly adapt the platform to address emerging applications?**
  - Focusing on your value-add rather than basic integration.

STMicroelectronics Corporate

STMicroelectronics: A Global Semiconductor Company



- **Q106 sales = $2.36B**
- **Approximately 50,000 employees**

*India, Africa, Latin America, Middle East

Global manufacturing infrastructure

ST Targeted Markets

Communications

Computer Peripherals

Digital Consumer

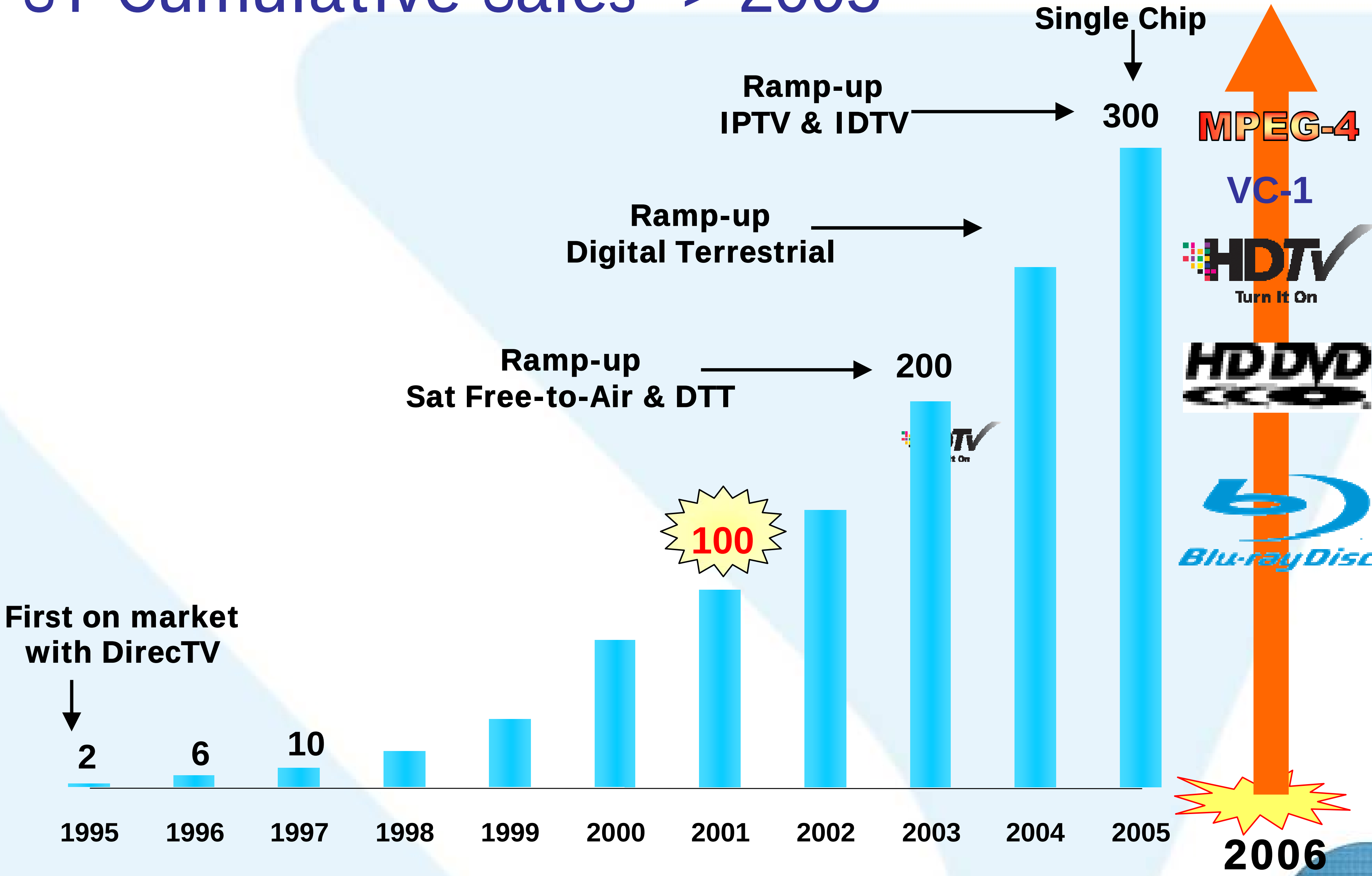
Automotive

Smart Cards

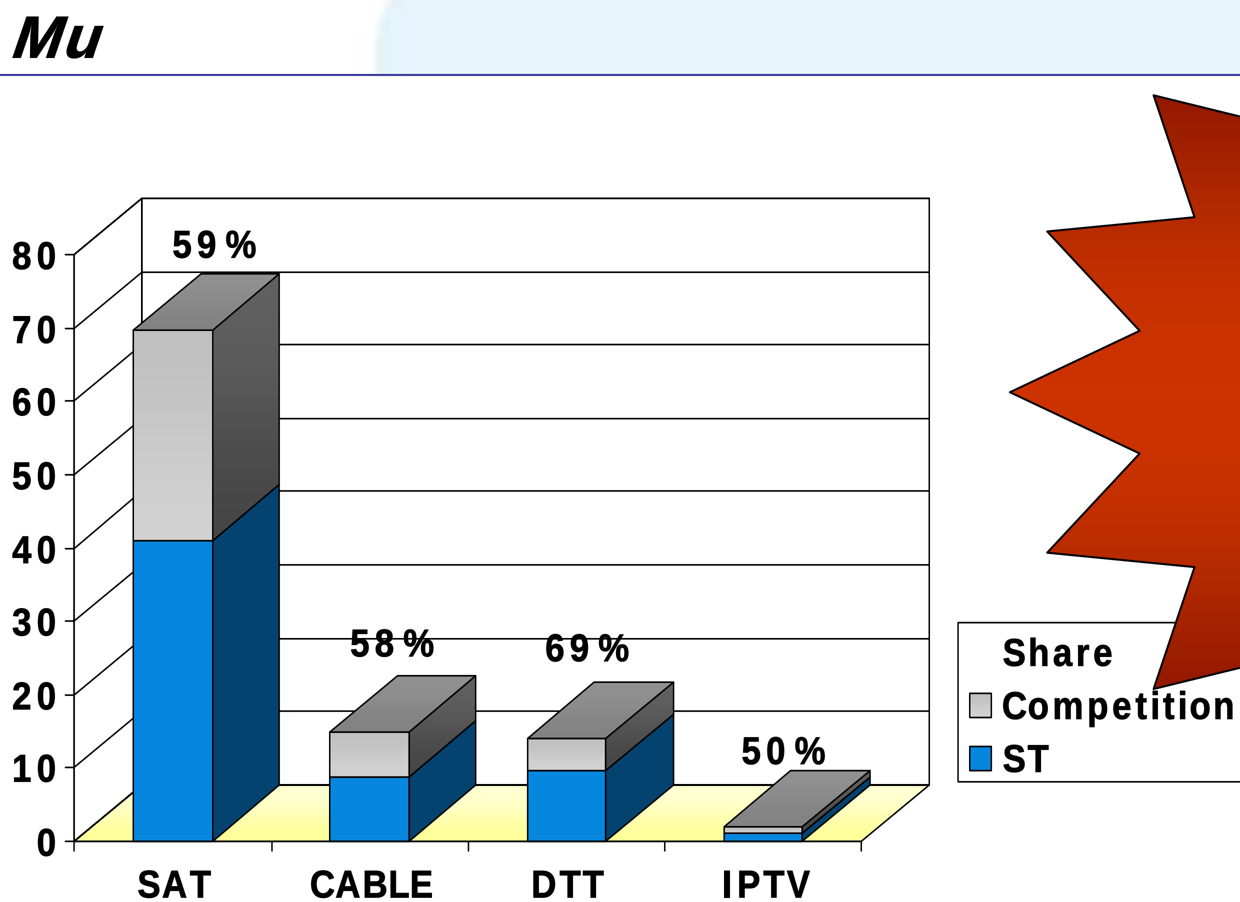
Key Applications:

Communications	Computer Peripherals	Digital Consumer	Automotive	Smart Cards
Wireless	Data storage	Set-Top Boxes	Engine/body/safety	Telephone
- Connectivity	Printers	DVDs	Car radio	Banking
- Mobile Phone	Imaging	Digital TVs	Car multimedia	User ID
- Portable MM	Monitors & Displays	Digital Cameras	Telematics	Security
Networking	Optical mouse	Digital Audio		

Enjoying the Digital Markets
ST Cumulative Sales -> 2005



2005: ST Shipped 60+ Million MPEG2 & 4 ICs



Total 2005 Market estimated at 100Mu

Market Share in MPEG STB market: 60%
 # 1 in Terrestrial
 # 1 in Satellite
 # 1 in Cable
 # 1 in IP

Source : ST Sales & Market Research, Feb. 2006

ST Strategic Development Axis

Drive advanced codec cost to mass deployment
- *Low cost architecture*
- *Latest process, single chip from beginning*
- *BOM integration*

Connectivity / IPTV
- *Server and client solutions*
- *Microsoft technologies*
- *Multi DRM / security*
- *Multimedia over IP*

Best in class HD video
- *Multicodec high performance*
- *Leverage decoder flexibility*
- *Image quality improvements*

Full platform offer
- *Multiple OS (Windows CE, OS21, Linux)*
- *Middleware community*
- *STB, HD-DVD/Blu-Ray, IPTV, TV*

Power consumption
- *Portable STB and DVD*
- *Environmental regulation*
- *Better reliability (no heat sink)*

The STi7109
Low cost HD STB SoC

STi7109 System on Chip Capabilities

High Definition Performance

- HD and SD DAC’s
  - Dual display composition
- YPbPr, DVI, HDMI output
- Graphics engine/blitter
- 266 MHz SH4-202 CPU core

Advanced codec and security

- Microsoft VC-1
- H264/AVC HP@L4
- MPEG2 MP@HL
- Windows Media DRM
- Secure Video Processor (SVP)

Connectivity

- Transport stream merger/router
  - Full range of front-end inputs
- Programmable transport stream interface (PTI)
- Digital video & audio inputs
- 10/100 Ethernet MAC
- USB 2.0 host controller
- SATA HDD interface for DVR
- IR in/out, UHF in for remote

Alas, no 1080p (but wait for our next-generation decoder!)

ST technology : Advanced Codecs

Flexible hardware + VLIW DSP architecture

- Single chip VC-1/H264 system fits easily in **32 bit DDR 200 MHz**
- **Pure DSP approach not appropriate** for SD and upper resolution
- Allows the cost point and power dissipation necessary for true CE devices

Status:

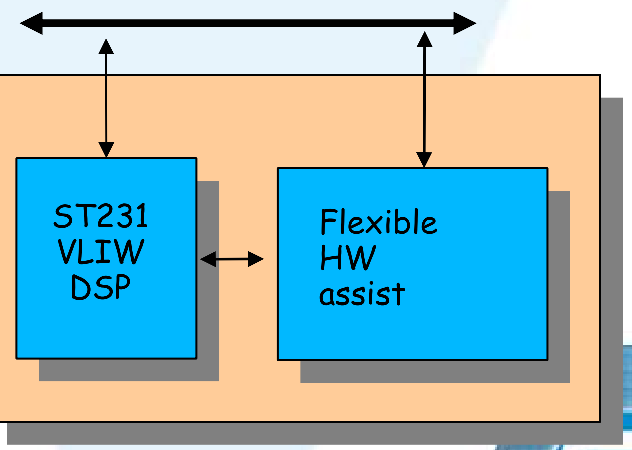
- AVC and VC-1 1080i single chip in production since 2005

Decoder capability (not exhaustive)

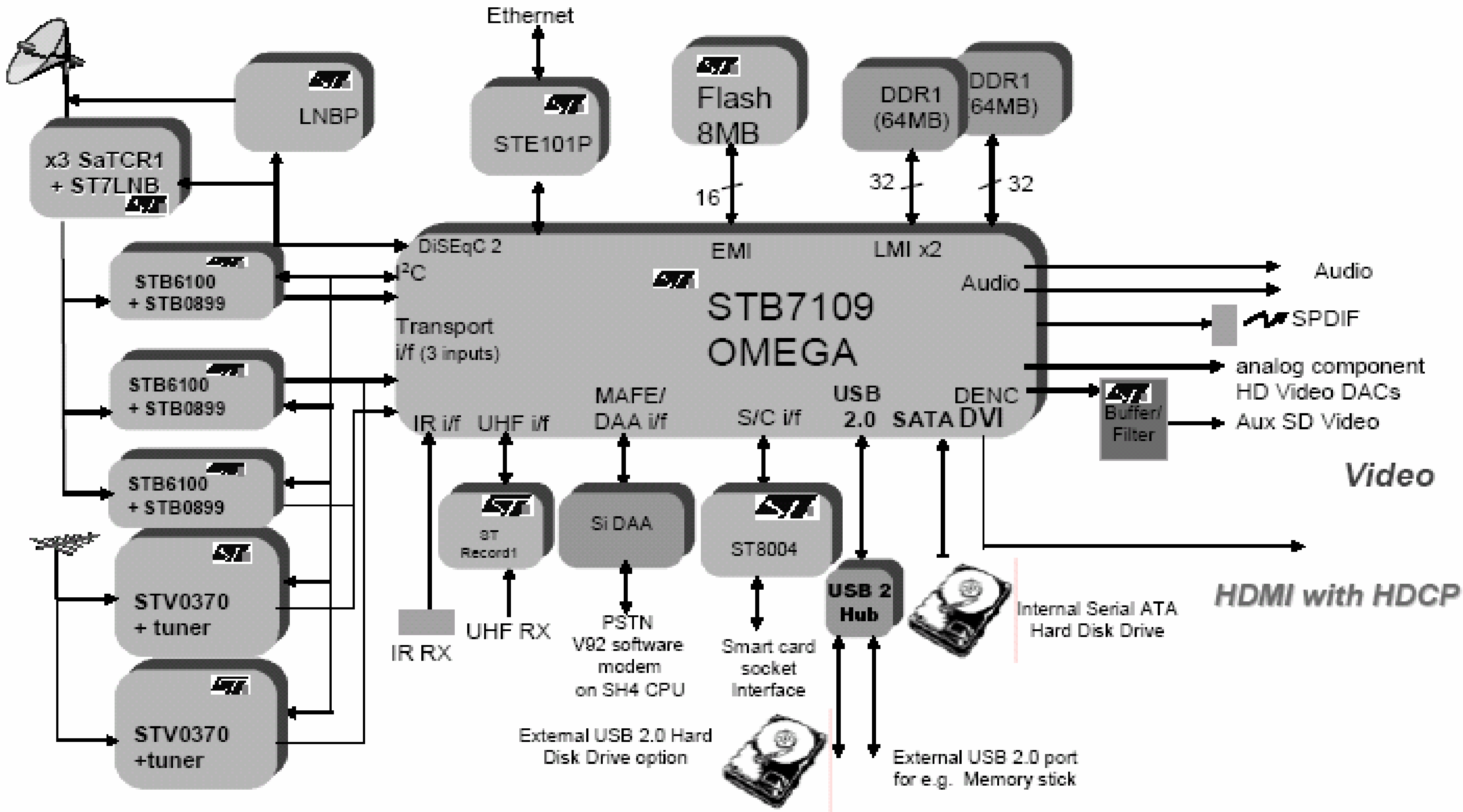
- MPEG2 MP@HL (1080i)
- **VC-1** Main and advanced profile level 3 (1080i)
- MPEG4-AVC Main and High profile Level 4 (**H.264 1080i**)
  - Baseline profile compatible with Main
- Many others (MPEG4 P2, DivX, H263…)

Field Software upgrades on going:

- Videoconferencing
- Transcoding
- Deblocking MPEG2



Hybrid IP STB Block Diagram



STB7109-Ref Reference Board Features

Major Parts

- STi7109
- 128 Mbytes DDR for OS, 64MBytes DDR for Video
- STE101P Ethernet PHY
- 8 Mbytes Flash
- SMSC LAN91C111 10/100 Ethernet LAN (for KITL interface)

Connectors

- 2 NIM Tuner Interfaces (297/299/360)
- ATAPI interface
- Video output : HDMI, 6xRCA (SD and HD), SCART (RGB, CVBS)
- Audio output : 4xRCA (internal & external DAC), Optical SPDIF
- Peripherals : 2 x RS232, IR Blaster input, SATA, USB host,
- JTAG connector for ST40 debugging via DCU

Power and PCB

- All power supplies derived from PC ATX PSU
- 4 layers main PCB design

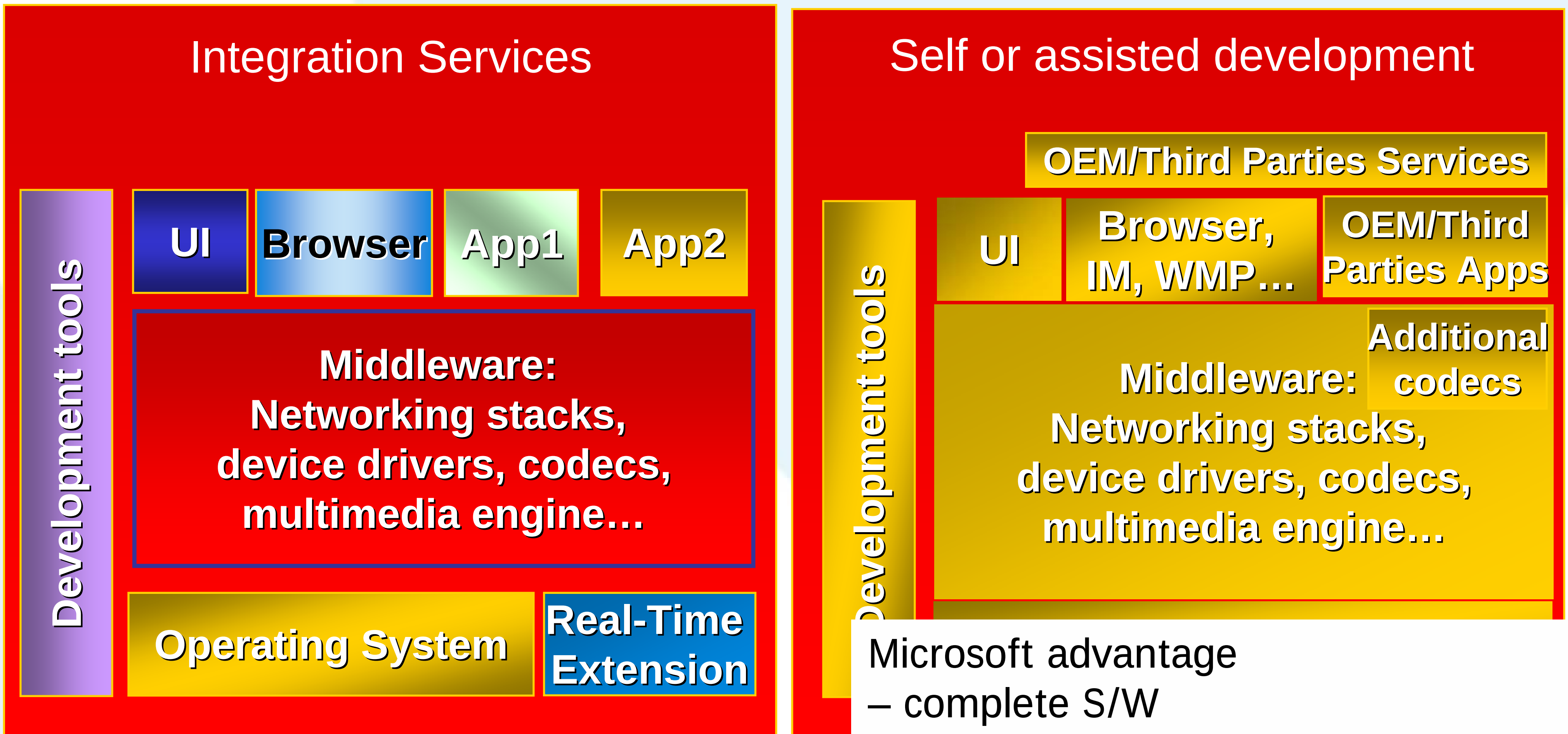
Windows CE
Multimedia BSP
for STi7109

Why choose Windows CE?

- **ST also offers Linux and OS21 for STi7109**
  - OS21 (ST proprietary OS) is great for simpler applications but doesn't have the connectivity protocols to keep up in our connected world
  - Linux can provide the connectivity protocols, but requires integrating the various pieces needed from different vendors.
- **With Windows CE, you let Microsoft worry about the standards "alphabet soup"**
  - WMDRM-ND, WMC, P4S, TCP/IP
- **One stop shop for licensing**
- **Strong toolset with Platform Builder to create, debug, and tune your system.**
- **Continuous improvement**
  - Monthly updates, Windows CE 6.0

Classic and Windows CE integration model

Classic Solution | **Windows CE**



Microsoft advantage
– complete S/W
Solution in House

Don't duplicate. Innovate !

ST and Microsoft: Better together

- **Microsoft makes great operating systems, but needs hardware to run it on.**
- **ST makes great chips, but doesn’t want to reinvent the wheel**
- **Starting in early 2005, the two companies began a stepwise program of joint development that has so far resulted in two BSP’s:**
  - Core BSP + Peripheral Drivers
  - Full Multimedia BSP taking full advantage of Microsoft’s recently launched Windows CE 5.0 Networked Media Device Feature Pack

STi7109 Network Media Device Demo

- **Windows CE Desktop**
  - CE Media Player
    - From SATA drive
    - From Flash Drive
- **VC-1 STAPI demo**
- **NMD Feature Pack user interface**
  - Audio playlists
  - Slideshow
  - MPEG2 HD streaming
    - 720p, 15Mbit/s
- **Alas, HD output needs HD projection, so this demo is SD output**

Core BSP

- **Main components**
  - OAL
  - Bootloader
  - Peripheral drivers
- **Developed and tested by Microsoft for MSTV program**
  - Architected for more general usage beyond MSTV
- **Released and maintained by ST**
- **Supported ST platforms:**
  - MB411
  - 7109-Ref
- **Available now**

7109 Peripheral Drivers in Core BSP

- **SATA**
- **7109 NDIS Ethernet**
- **USB 2.0 Host (OHCI+EHCI)**
  - Mass storage
  - HID
- **IR Receiver**
  - RCMM protocol driver in .DLL format
- **Serial UART**

Multimedia BSP Components

- **Multimedia BSP includes core BSP plus:**
- **Transport Stream Demux/splitter**
  - ASF wrapping VC-1
  - MPEG2 TS wrapping H.264, MPEG-2
- **GDI display driver with acceleration**
- **DirectShow Video Renderer with acceleration**
  - VC-1/WMV9, AVC/H.264, MPEG2
- **DirectShow Audio Renderer with acceleration**
  - WMA/Pro, MP3, AC3
  - Wave audio driver for system sounds
- **Video outputs on HDMI/DVI and YPbPr**
  - 480i, 480p, 576i, 576p, 720p, 1080i resolutions
- **WMDRM-ND (Cardea)**
- **Networked Media Device Feature Pack implementation**
- **Production release late Q406**

STAPI on Windows CE

- **STAPI is ST's proprietary API for video and graphics**
- **STAPI functionality available in two ways**

1. **Static library linked to the application**
    - Minimizes porting effort for existing STAPI applications
2. **Wrapped by DirectShow and GDI drivers to expose standard Windows CE API's.**

- **STAPI not necessary for Core BSP porting work**

Windows CE 6.0 status

- **Initial multimedia development started on 5.0**
- **After each milestone, 6.0 BSP brought to same level of functionality as 5.0 BSP**
- **Production release of Multimedia BSP likely to be 6.0 only**
  - Launch customer requirements will dictate

BSP Licensing

For Development:

- **Core BSP requires license agreement**
  - Similar terms to Windows CE EULA with derivative and redistribution rights
  - No charge for this license
- **Multimedia BSP requires extended license agreement, and STAPI license agreement**
  - STAPI license maintains one time USD $10,000 license fee
  - No additional charge for existing STAPI licensees
- **No licensing from Microsoft beyond standard Platform Builder EULA**

For Production:

- **No additional licensing from ST**
- **OEM Distribution license required from Microsoft**

Multimedia BSP vs. MSTV Client

- **ST and Microsoft are well along a joint development path to release an MSTV Client running on a STi7109-based platform from Scientific Atlanta**
- **Core BSP became the starting point for both the MSTV program and the Multimedia BSP**
- **Neither the Core BSP nor the Multimedia BSP include any MSTV client middleware**
- **Porting the Core BSP to an OEM platform is a good first step in preparation for a potential MSTV offering or the Multimedia BSP**
  - ST cannot provide any MSTV client components, only a compatible BSP.

Application Space for STi7109/Windows CE

- **IP Set top Box**
  - Pure IP STB, or hybrid STB with terrestrial or satellite tuner
- **Digital Media Adaptor meeting PlaysForSure requirements**
- **Network Projector (New feature in Windows Vista)**
- **Connected DVD player**
  - HD-DVD will require future silicon versions
- **Media Center Extender (MCX)**
- **Portable DVD player**
- **Video Conferencing**
- **Unique combinations of the above, anything that needs that calls for IP in, HD out**

New functionality/price points

- **ST and Microsoft joint development creates the space for a sub-$200 retail price point for a High-Def Digital Media Adapter**
- **Compare that price point against:**
  - Low end PC: $600
  - Xbox360: $399
  - V1.0 Media Center Extender: $239
- **Allow consumers to fill that need to have more HD content on their pretty new flat screen**

Recipe for HD Connectivity

- Start with the DirectShow filters exposing the powerful and flexible 7109 decoder and audio/video renders
- Fold in the NDIS driver for the 7109 bus master Ethernet block with the Windows CE TCP/IP stack
- Add two parts HTTP and the UPnP A/V framework, from the Windows CE catalog
- Mix in Windows Media Connect 2.0 found in the free Network Media Device Feature Pack download
- Dash with premium content via Windows Media DRM for Networked Devices (Cardea)
- Bake with PlaysForSure 2.0 logo for customer confidence
- Quickly serve your own innovative combination of functions, while it is still hot

I want it, where can I get it?

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Q and A

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